

APPENDICES

Experiment 1: Quinine Analysis by Fluorescence Spectroscopy

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APPENDIX A: Calibration Curves

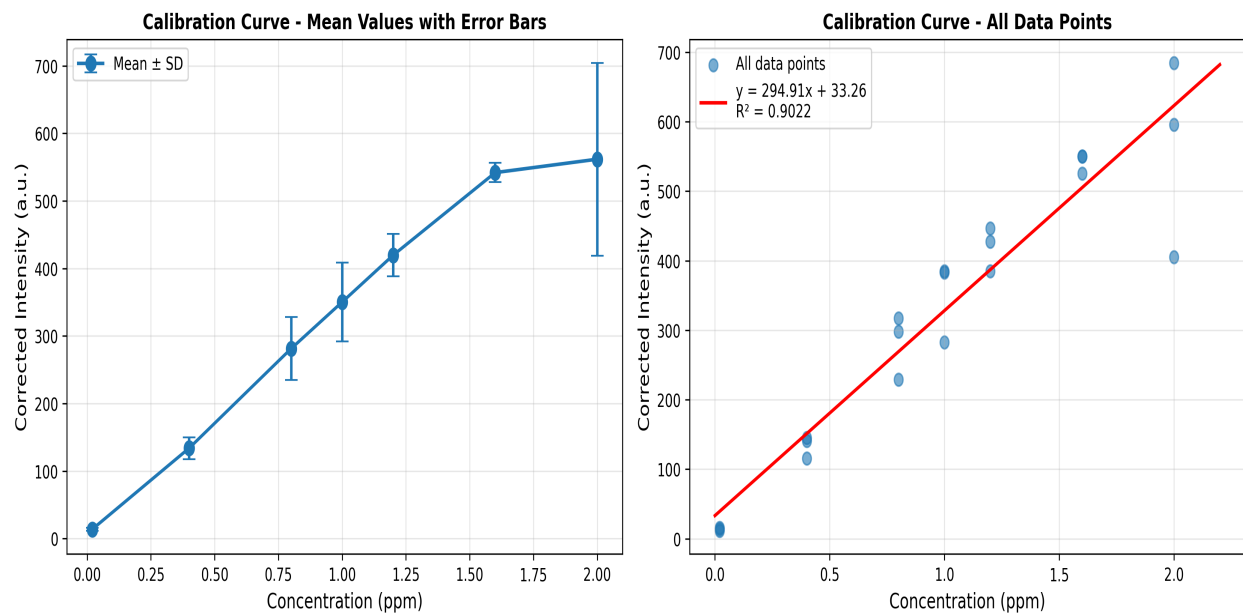


Figure A1: Calibration curves for quinine fluorescence. Left panel shows mean intensity values with standard deviation error bars. Right panel shows all 21 data points with linear regression line ($y = 294.91x + 33.26$, $R^2 = 0.9022$).

APPENDIX B: Complete Data Tables

B.1 Calibration Standards Data

Concentration (ppm)	Intensity 1	Intensity 2	Intensity 3	Mean	SD	SE
0.0200	14.2093	11.7838	15.6252	13.8728	1.9427	1.1216
0.4000	115.5347	145.5005	140.9392	133.9915	16.1460	9.3219
0.8000	228.8266	317.2895	298.0940	281.4033	46.5334	26.8661
1.0000	282.7963	385.1652	383.0237	350.3284	58.4943	33.7717
1.2000	385.1930	427.2684	446.6821	419.7145	31.4329	18.1478
1.6000	550.1447	525.4876	550.1760	541.9361	14.2449	8.2243
2.0000	405.2259	595.5215	684.6166	561.7880	142.7174	82.3979

B.2 Regression Statistics

Parameter	Value
Slope (m)	294.905970
Intercept (b)	33.256381
R ²	0.902202
Std Error (sy)	64.100658
Slope StdErr (sm)	22.275135
Intercept StdErr (sb)	26.356833
LOD (ppm)	0.717287
LOQ (ppm)	2.173597

B.3 Tonic Water Sample Results

Sample	Intensity	Concentration (ppm)	Uncertainty (ppm)
Tci	260.93	0.7720	0.2232
Ti	221.16	0.6372	0.2242
Tn	270.81	0.8055	0.2230

APPENDIX C: Sample Calculations

C.1 Calculation of Concentration from Calibration Curve

Given calibration equation: $y = 294.91x + 33.26$

For sample Tci with corrected intensity $y = 260.93$ a.u.

Solving for x (concentration):

$$x = (y - b) / m$$

$$x = (260.93 - 33.26) / 294.91$$

$$x = 227.67 / 294.91$$

$$x = 0.772 \text{ ppm}$$

C.2 Calculation of Uncertainty

Using the formula from Harris (8th Ed.):

$$s_x = (s_y / |m|) \times \sqrt{[(1/k) + (1/n) + ((y - \bar{y})^2) / (m^2 \times \sum (x_i - \bar{x})^2)]}$$

Where:

- $s_y = 64.10$ (standard error of estimate)
- $m = 294.91$ (slope)
- $k = 1$ (single measurement)
- $n = 21$ (calibration points)
- $y = 260.93$ (measured intensity)
- $\bar{y}$ = mean of calibration intensities

Result: $s_x \approx 0.22$ ppm

C.3 LOD and LOQ Calculations

$$\text{LOD} = 3.3 \times s_y / |m| = 3.3 \times 64.10 / 294.91 = 0.717 \text{ ppm}$$

$$\text{LOQ} = 10 \times s_y / |m| = 10 \times 64.10 / 294.91 = 2.174 \text{ ppm}$$